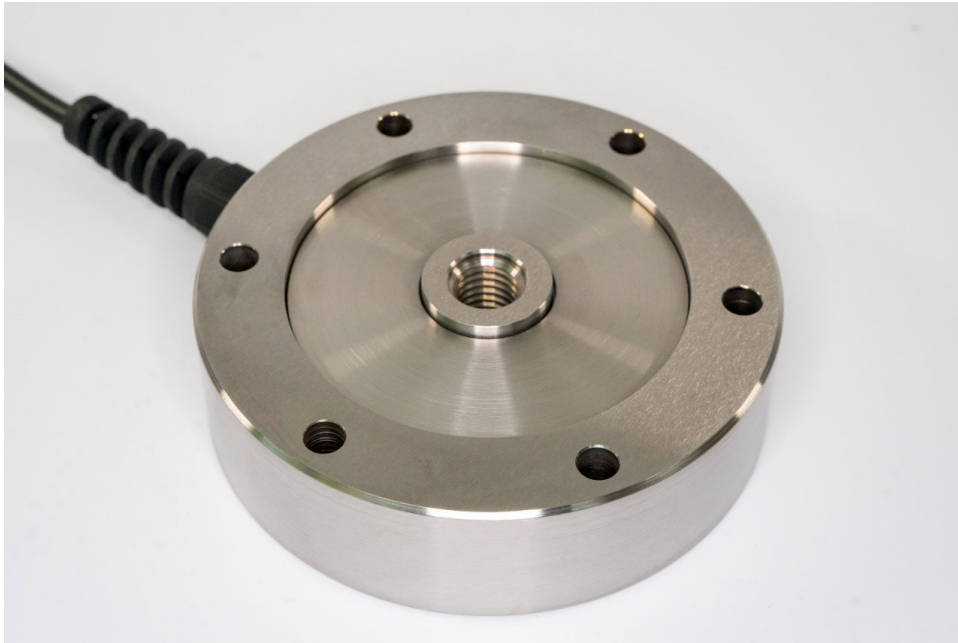




F252 Pancake Loadcell

Geometry: Low profile axial loadcell for use in force measurements in tension and compression.

With bi-directional versions there is a small difference between the output signal for compression and tension. All standard bi-directional loadcells are calibrated in both modes and the output for each direction is stated on the test / calibration certificate.

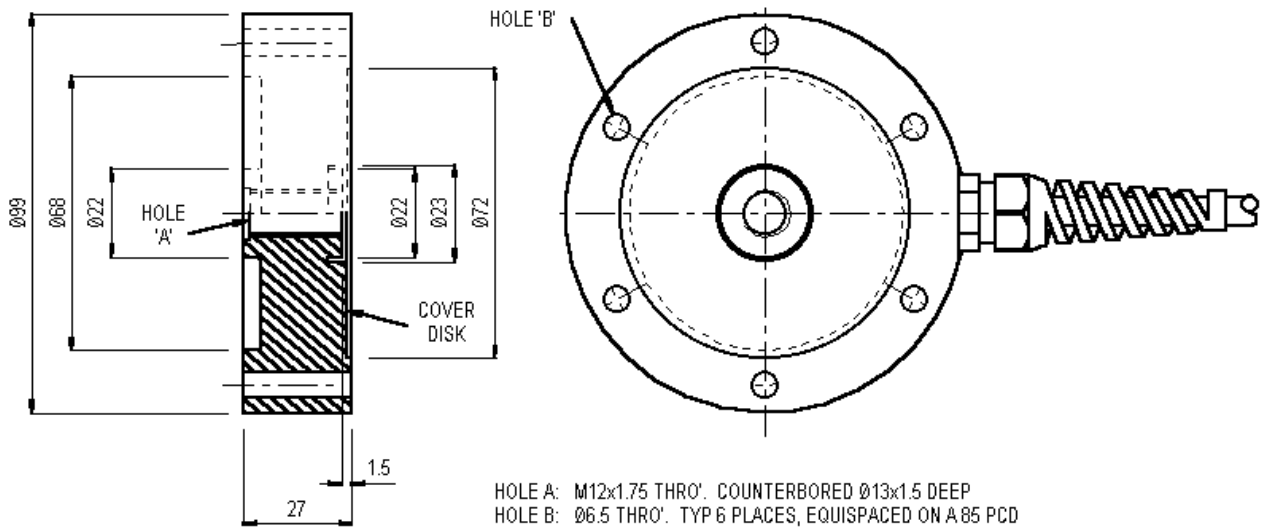
The F252 is ideal for engineering force measurements particularly in applications where there is a limit on the height of the loadcell. It can be used for test machines and a wide range of general industrial applications.

We are happy to design variants of this loadcell to meet your specific requirements. Versions can be manufactured for fully compensated operation up to +250°C. Please consult our engineering department.

Features

- ✓ Tension / compression / bi-directional calibration
- ✓ Compact size
- ✓ Low deflection
- ✓ Hardened stainless steel body
- ✓ Traceable calibration with certificate included in the standard price
- ✓ Standard 1 year warranty

Outline



Specifications

Parameter	Value	Unit
Non-linearity - Terminal	± 0.1	% RL
Hysteresis	± 0.1	% RL
Creep - 20 minutes	± 0.05	% AL
Repeatability	± 0.02	% RL
Rated output - Nominal	1.6	mV/V
Rated output - Rationalised	1.5	mV/V
Rationalisation tolerance (applies to single direction calibrations)	± 0.5	% RL
Zero load output	± 4	% RL
Temperature effect on rated output per °C	± 0.005	% AL
Temperature effect on zero load output per °C	± 0.02	% RL
Temperature range - Compensated	-10 to +50	°C
Temperature range - Safe	-10 to +80	°C
Excitation voltage - Recommended	10	V
Excitation voltage - Maximum	20	V
Bridge resistance up to 400kgf	350	Ω
above 400kgf	700	Ω
Insulation resistance - Minimum at 50Vdc	500	M Ω
Overload - Safe	50	% RL
Overload - Ultimate	100	% RL
Weight - Nominal (excluding cable)	840 to 940	g

Order Codes

Code	Description
F252CF00K0	Compression, unrationalised
F252TF00K0	Tension, unrationalised
F252UF00K0	Bi-directional, unrationalised
F252CF00KN	Compression, rationalised
F252TF00KN	Tension, rationalised
F252UF00KN	Bi-directional, rationalised

Structural Stiffness

Range (kN)	Stiffness (N/m)
1	3.0×10^6
2	6.0×10^6
4	1.2×10^7
8	2.4×10^7
10	3.0×10^7

Notes

- AL = Applied load.
- RL = Rated load.
- Temperature coefficients apply over the compensated range.
- The load must be applied directly through the central loading axis.

Connections

The loadcell is fitted with 2 metres of PVC insulated 4 core screened cable type 7-2-4C.

Excitation + = Red, Excitation - = Blue, Signal + = Yellow, Signal - = Green, Screen = Orange.

Reverse the signal connections to obtain a positive signal in tension mode. The screen is not connected to the loadcell body.